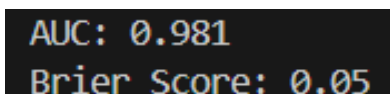


Cainan Squibb

Catcher Framing Project

My process for beginning this project was to aggregate catcher data first. I did this because I thought the catcher's framing value may help my model make better predictions. To create the new_output CSV, I counted the total number of opportunities each catcher had per year as well as the total number of strikes called when they were catching. To find the total number of strikes added, I checked whether the pitch was called a ball or strike and if it was called a strike, I checked whether the location was outside the bound of the strike zone provided by TOP_ZONE, BOT_ZONE, and $\pm 8.5/12$ (half the distance across Homeplate converted to feet). If the pitch met these criteria, I incremented CALLED_STRIKES_ADDED. However, if the pitch was within the bounds set and was called a ball, I decremented CALLED_STRIKES_ADDED. After CALLED_STRIKES_ADDED is calculated, I divide it by OPPORTUNITIES and multiply it by 100 to get the CALLED_STRIKES_ADDED_PER_100.

To predict the probability of a pitch being called a strike, I used a Random Forest Classification model. I used a maximum depth of 20 and 1000 estimators for my initial model. I started by deciding what features to include in my model. Although I did not think pitch type would be significant, I still chose to use the pitch types initially. This required me to encode them using a label encoder. Besides the two pitch type variables, I used the following variables in my initial model: PLATELOCHEIGHT, PLATELOCSIDE, BOT_ZONE, TOP_ZONE, INDUCEDVERTBREAK, HORZBREAK, UMPIRE_ID, CALLED_STRIKES_ADDED_PER_100. These were variables that I believed to have an impact on an umpire's ability to call the pitch accurately. After I finished my model, I tested it using AUC and Brier scores:



```
AUC: 0.981
Brier Score: 0.05
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Figure 1: Initial AUC & Brier Scores

As seen by *Figure 1*, the model is very accurate when predicting balls and strikes. To understand my model further, I ran feature and permutation importance tests.



Figure 2: Feature & Permutation Importance

As seen by *Figure 2*, PLATELOCHEIGHT and PLATELOCSIDE dominate the model's classification, which makes sense because the location is the most pertinent metric for determining balls and strikes. Because the model predominantly predicts using two variables, in my final model only PLATELOCHEIGHT, PLATELOCSIDE, BOT_ZONE, TOP_ZONE are used. I kept BOT_ZONE and TOP_ZONE due to its relation to the pitch location which I used to confirm my model's accuracy manually. After modifying my model's features, I used grid searching to determine the optimal depth and estimators. The grid search produced the optimal maximum depth and estimator values of 10 and 1200. These values produced the AUC and Brier scores shown in *Figure 3* below.

AUC: 0.982
Brier Score: 0.049

Figure 3: Final AUC & Brier Scores